

Advanced Java MCQ 3

1. Which method is called by the server to initialize a servlet?

- A. `service()`
- B. `init()`
- C. `start()`
- D. `constructor()`

Answer: B

The `init()` method is called once when the servlet is first loaded to perform any initialization tasks.

2. What is the correct file extension for a JSP file?

- A. `.html`
- B. `.jsp`
- C. `.java`
- D. `.xml`

Answer: B

JSP files use the `.jsp` extension.

3. Which of the following is true about the `doGet()` method?

- A. It handles POST requests
- B. It is called when a GET request is made
- C. It is never used in servlets
- D. It closes the servlet

Answer: B

The `doGet()` method is invoked by the server when a GET request is received.

4. In JSP, what does the `<%= expression %>` tag do?

- A. Includes a directive
- B. Declares a variable

- C. Outputs the result of the expression
- D. Creates a comment

Answer: C

`<%= expression %>` evaluates and outputs the result directly into the response.

5. Which object is used to access session data in a servlet?

- A. `ServletConfig`
- B. `ServletRequest`
- C. `HttpSession`
- D. `ServletContext`

Answer: C

`HttpSession` is used for session tracking and storing user data.

6. Which method is used to forward a request to another resource in a servlet?

- A. `sendRedirect()`
- B. `forward()`
- C. `include()`
- D. `dispatch()`

Answer: B

The `RequestDispatcher.forward()` method forwards a request internally on the server.

7. Which of the following statements about `sendRedirect()` is true?

- A. It is handled entirely on the server
- B. It keeps request attributes intact
- C. It tells the client to issue a new request
- D. It does not change the URL in the browser

Answer: C

`sendRedirect()` causes the client to make a new request to a different URL.

8. What happens if you call `session.invalidate()` in a servlet?

- A. The session timeout is extended
- B. The session data is reset
- C. The session is destroyed
- D. The user is logged in

Answer: C

`invalidate()` method destroys the session and removes all attributes.

9. Filters in servlets are mainly used for:

- A. Database connection
- B. Compiling servlets
- C. Preprocessing and postprocessing requests
- D. Parsing JSP

Answer: C

Filters intercept requests before they reach the servlet, useful for logging, authentication, etc.

10. Which interface must a class implement to define a filter in Java EE?

- A. `HttpServlet`
- B. `ServletContextListener`
- C. `Filter`
- D. `SessionListener`

Answer: C

A class must implement `javax.servlet.Filter` to become a filter.

11. What is the correct lifecycle order of a filter?

- A. `init()`, `destroy()`, `doFilter()`
- B. `doFilter()`, `init()`, `destroy()`
- C. `init()`, `doFilter()`, `destroy()`
- D. `doFilter()`, `destroy()`, `init()`

Answer: C

Filters follow `init()`, then `doFilter()` for each request, and `destroy()` at the end.

12. What does `isAsyncSupported=true` enable in a servlet or filter?

- A. Synchronous execution
- B. Blocking mode
- C. Asynchronous processing
- D. Fast compiling

Answer: C

It enables asynchronous processing, allowing servlets to process requests on a different thread.

13. Which class is used to start asynchronous processing in a servlet?

- A. `AsyncHandler`
- B. `AsyncServlet`
- C. `AsyncContext`
- D. `AsyncManager`

Answer: C

`AsyncContext` manages asynchronous processing in servlets.

14. Which method starts async processing in `HttpServletRequest`?

- A. `beginAsync()`
- B. `runAsync()`
- C. `startAsync()`
- D. `initAsync()`

Answer: C

`startAsync()` begins asynchronous processing.

15. In JSP, which scope is NOT valid for a bean?

- A. `application`
- B. `session`
- C. `request`
- D. `global`

Answer: D

The valid scopes in JSP are page, request, session, and application.

16. How can you prevent session fixation attacks in servlets?

- A. Use cookies
- B. Avoid using sessions
- C. Regenerate session ID on login
- D. Use GET parameters

Answer: C

To prevent session fixation, regenerate session IDs after authentication.

17. Which of these is a secure way to store sensitive data in web applications?

- A. Store in plain text in cookies
- B. Use HTTPS and server-side sessions
- C. Use GET parameters
- D. Store in local storage

Answer: B

HTTPS with server-side sessions helps protect sensitive data.

18. What is the default session timeout in most Java web containers?

- A. 1 minute
- B. 10 minutes
- C. 30 minutes
- D. 60 minutes

Answer: C

The default session timeout is usually 30 minutes unless configured otherwise.

19. Which of the following is used to define filters in a Java EE web app?

- A. web.xml
- B. filter.xml

- C. `server.xml`
- D. `context.xml`

Answer: A

Filters are defined in `web.xml` or via annotations.

20. In JSP, which tag includes another JSP at request time?

- A. `<jsp:forward>`
- B. `<jsp:useBean>`
- C. `<jsp:include>`
- D. `<jsp:param>`

Answer: C

`<jsp:include>` includes a file at request time.

21. Which header can be used to protect against XSS attacks?

- A. `X-Auth-Token`
- B. `X-Content-Type-Options`
- C. `Content-Security-Policy`
- D. `X-Download-Options`

Answer: C

`Content-Security-Policy` restricts what content the browser is allowed to execute.

22. Which response code indicates a redirect?

- A. 200
- B. 404
- C. 302
- D. 500

Answer: C

HTTP 302 indicates a temporary redirect.

23. How do you store user data in a session from a servlet?

- A. `session.put()`
- B. `request.setAttribute()`
- C. `session.setAttribute()`
- D. `context.setAttribute()`

Answer: C

`session.setAttribute()` stores data in the session scope.

24. When is the `destroy()` method of a servlet called?

- A. After every request
- B. On servlet initialization
- C. When server shuts down or servlet is unloaded
- D. When a new request is received

Answer: C

`destroy()` is called once when the servlet is being taken out of service.

25. Which of the following is NOT a valid redirect URL pattern?

- A. `/home.jsp`
- B. `https://example.com/dashboard`
- C. `dashboard.jsp`
- D. `file://local/file.jsp`

Answer: D

Redirecting to `file://` URLs is insecure and not permitted by browsers for web responses.

26. Which of the following objects is NOT implicitly available in a JSP page?

- A. `request`
- B. `response`
- C. `config`
- D. `servlet`

Answer: D

`servlet` is not an implicit object. JSP provides objects like `request`, `response`, `session`, `application`, etc.

27. How can you configure the session timeout in a web application?

- A. In `web.xml` using `<session-timeout>`
- B. In `server.xml`
- C. Through a JSP tag
- D. Using a filter

Answer: A

The `<session-timeout>` element in `web.xml` defines the default timeout for sessions.

28. What will happen if `getWriter()` and `getOutputStream()` are used together in a servlet?

- A. Output is doubled
- B. Servlet will crash
- C. `IllegalStateException` is thrown
- D. It works fine

Answer: C

You cannot use both methods; doing so throws an `IllegalStateException`.

29. What is the role of `RequestDispatcher.include()`?

- A. Forwards the request and terminates
- B. Redirects client to another page
- C. Includes response from another resource
- D. Deletes previous session

Answer: C

`include()` includes the content of another resource in the current response.

30. Which tag is used in JSP to declare variables or methods?

- A. `<%! %>`
- B. `<%= %>`
- C. `<%-- --%>`
- D. `<%@ %>`

Answer: A

`<%! %>` is used to declare methods or variables at the class level.

31. What is a key advantage of asynchronous servlets?

- A. They simplify code
- B. They allow blocking operations
- C. They handle long-running tasks without blocking threads
- D. They support only GET requests

Answer: C

Asynchronous servlets improve performance by freeing threads during long operations.

32. Which annotation is used to define a servlet in modern Java EE applications?

- A. `@WebFilter`
- B. `@WebServlet`
- C. `@Servlet`
- D. `@ServletMapping`

Answer: B

`@WebServlet` is used to declare a servlet without editing `web.xml`.

33. Which of the following methods is used to retrieve a session without creating a new one?

- A. `request.getSession()`
- B. `request.getSession(false)`
- C. `session.get()`
- D. `request.getNewSession()`

Answer: B

`getSession(false)` returns the current session or null if none exists.

34. What is the purpose of the `@WebFilter` annotation?

- A. To declare a servlet
- B. To define security constraints
- C. To declare a filter without `web.xml`
- D. To create session tracking

Answer: C

`@WebFilter` is used to declare a filter directly in the class.

35. How can you prevent Cross-Site Request Forgery (CSRF) in web apps?

- A. Disable cookies
- B. Use strong passwords
- C. Use CSRF tokens in forms
- D. Minify JavaScript

Answer: C

CSRF tokens ensure that form submissions come from legitimate sources.

36. What is returned by `request.getAttribute("key")` if no attribute exists?

- A. null
- B. empty string
- C. 0
- D. throws exception

Answer: A

If the attribute doesn't exist, `getAttribute()` returns `null`.

37. Which status code is used for "Forbidden"?

- A. 200
- B. 301
- C. 403
- D. 500

Answer: C

HTTP 403 means the server understands the request but refuses to authorize it.

38. Which method of `HttpServletResponse` is used for redirection?

- A. `forward()`
- B. `sendError()`
- C. `sendRedirect()`
- D. `include()`

Answer: C

`sendRedirect()` instructs the browser to make a new request to another URL.

39. Which JSP directive is used to include another JSP at translation time?

- A. `<jsp:include>`
- B. `<%@ include file="..." %>`
- C. `<jsp:forward>`
- D. `<jsp:directive>`

Answer: B

`<%@ include %>` includes content during JSP translation, not runtime.

40. What is the purpose of `response.setHeader("Cache-Control", "no-store")`?

- A. To allow caching
- B. To force redirect
- C. To prevent caching
- D. To clear cookies

Answer: C

It prevents browsers from storing any part of the response.

41. In asynchronous servlets, what method is used to complete processing?

- A. `end()`
- B. `close()`
- C. `complete()`
- D. `finish()`

Answer: C

`AsyncContext.complete()` signals that async processing is done.

42. Which class is used to define filter configuration?

- A. `FilterChain`
- B. `FilterConfig`
- C. `RequestConfig`
- D. `ServletRequest`

Answer: B

`FilterConfig` allows access to filter initialization parameters.

43. What is the scope of variables declared using `<%! int x = 5; %>` in JSP?

- A. Page scope
- B. Request scope
- C. Instance variable (available across requests to that page instance)
- D. Session scope

Answer: C

Variables in `<%! %>` are declared at the servlet class level.

44. What is the function of the `FilterChain.doFilter()` method?

- A. Terminates filtering
- B. Sends the response to client
- C. Passes request/response to next filter or servlet
- D. Creates a new filter

Answer: C

It forwards the request and response down the filter chain.

45. Which of the following best prevents SQL Injection in Java servlets?

- A. Using String concatenation for SQL queries
- B. Validating user input length

- C. Using `PreparedStatement`
- D. Escaping input manually

Answer: C

`PreparedStatement` safely binds parameters and prevents SQL injection.